

# Supply Chain Management

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# Ch 10. Coordination in a Supply Chain

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# Chapter Overview

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1. Lack of Supply Chain Coordination and its Impact on Performance
2. Obstacles to Coordination in a Supply Chain
3. Managerial Levers to Improve Coordination
4. Some Practical Approaches to Improve Supply Chain Coordination



## 10.1. Lack of Supply Chain Coordination and its Impact on Performance

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- A lack of coordination occurs either because different stages of the supply chain have local objectives that conflict or because information between stages is delayed and distorted
- Supply chain coordination improves the efficiency and responsiveness of the supply chain
- The fundamental challenge today is for supply chain to achieve coordination in spite of multiple ownership and increased product variety



## 10.1. Lack of Supply Chain Coordination and its Impact on Performance

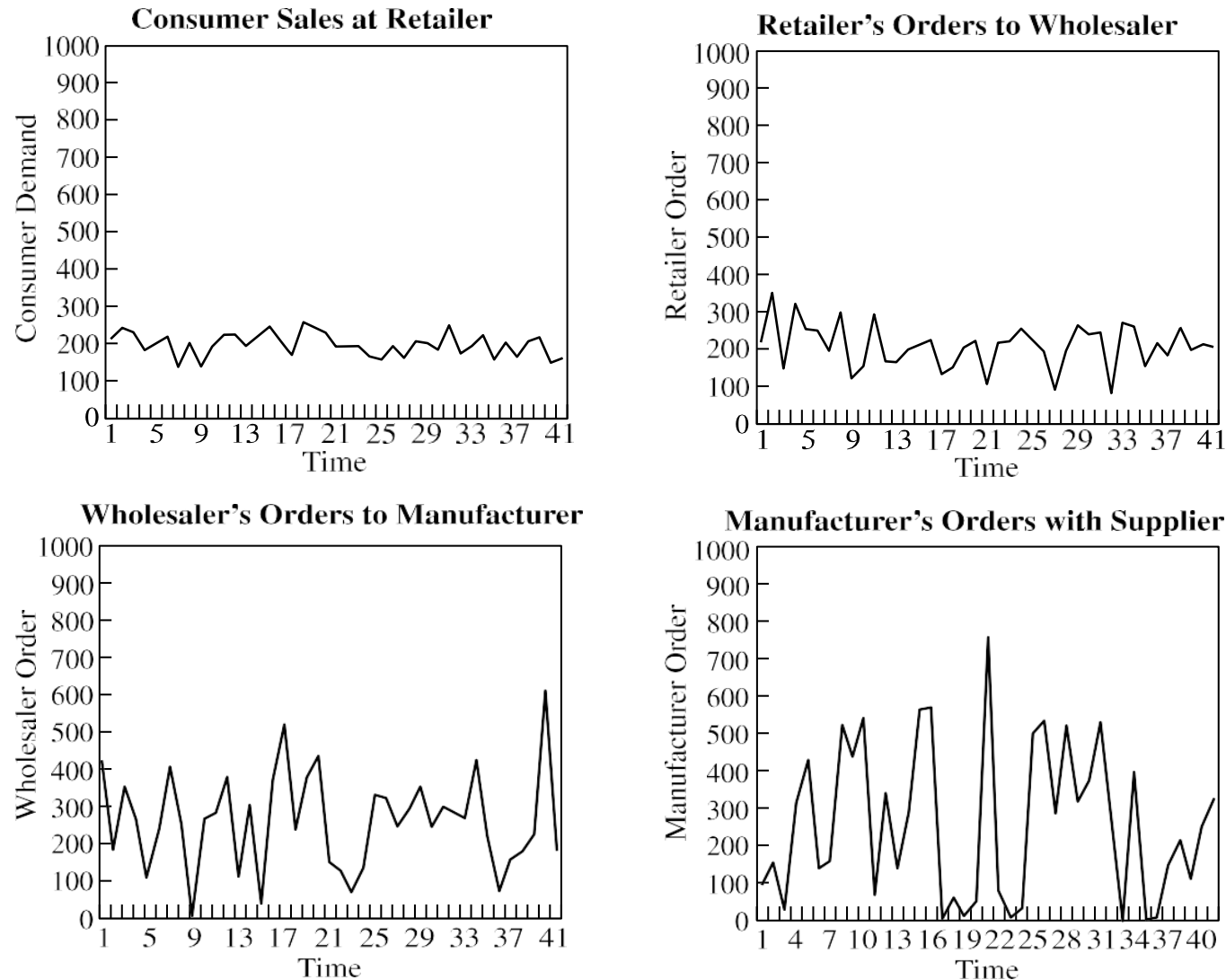
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- Bullwhip effect
  - One typical outcome of the lack of supply chain coordination is the bullwhip effect, in which fluctuations in orders increase as they move up the supply chain from retailers to wholesalers to manufacturers to suppliers
  - It distorts demand information within the supply chain, with each stage having a different estimate of what demand looks like
  - Observations in Procter & Gamble (P&G), Hewlett-Packard (HP), and apparel and grocery industries



# 10.1. Lack of Supply Chain Coordination and its Impact on Performance

- Demand fluctuation at different stages of a supply chain (Figure 10.1)



## 10.1. Lack of Supply Chain Coordination and its Impact on Performance

- Impact of the lack of coordination on supply chain performance

| Performance Measure           | Impact of the Lack of Coordination |
|-------------------------------|------------------------------------|
| Manufacturing cost            | Increases                          |
| Inventory cost                | Increases                          |
| Replenishment lead time       | Increases                          |
| Transportation cost           | Increases                          |
| Shipping and receiving cost   | Increases                          |
| Level of product availability | Decreases                          |
| Profitability                 | Decreases                          |



## 10.2. Obstacles to Coordination in a Supply Chain

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- Incentive Obstacles (1)
  - Local optimization within functions or stages
  - Sales force incentives
- Information Processing Obstacles (2)
  - Forecasting based on orders (not customer demand)
  - Lack of information sharing between stages



## 10.2. Obstacles to Coordination in a Supply Chain

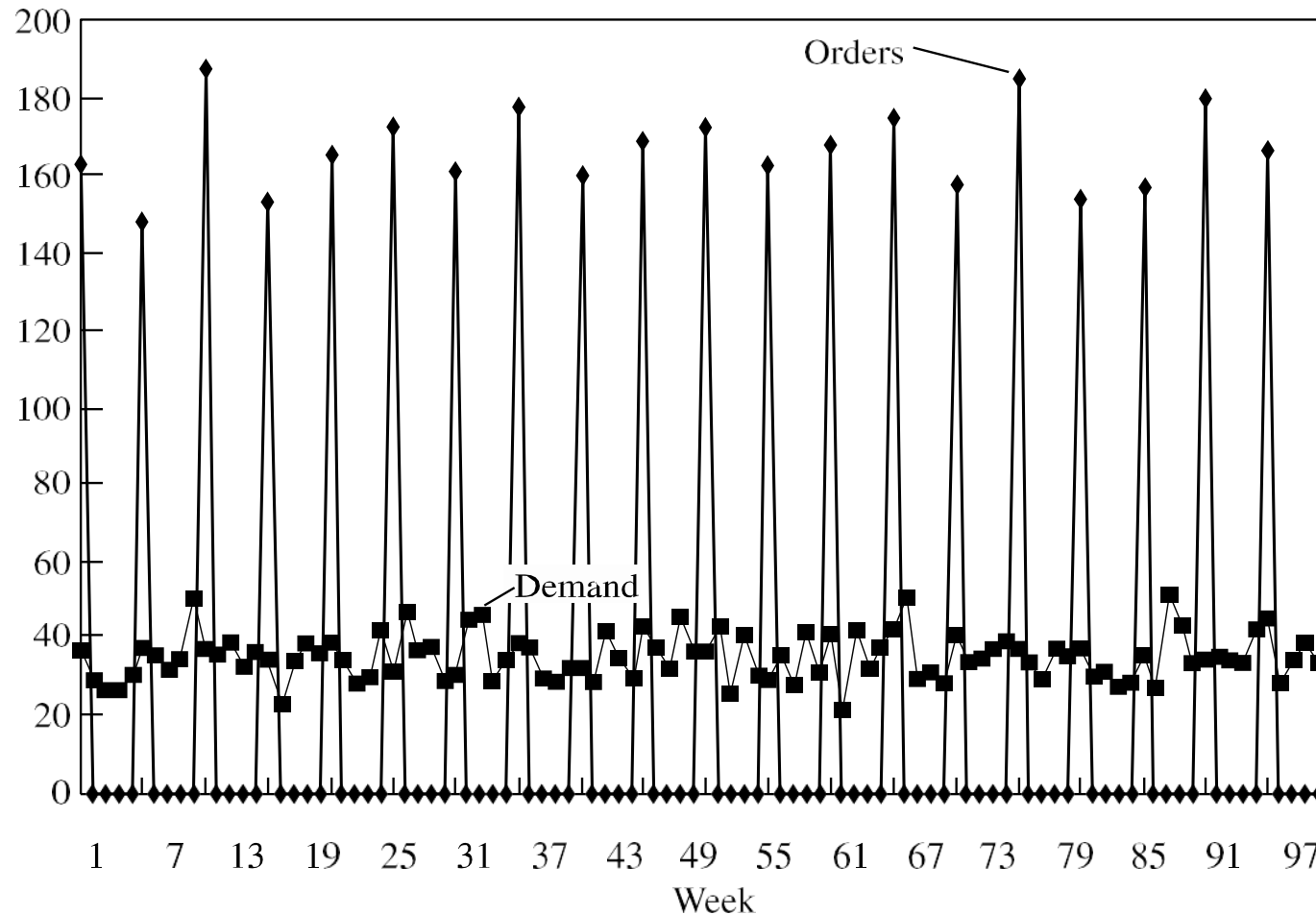
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- Operational Obstacles (3)
  - Ordering in large lots (Fig. 10.2)
  - Large replenishment lead times
  - Rationing and shortage gaming (i.e., supply shortages)
- Pricing Obstacles (4)
  - Lot-size-based quantity discounts
  - Price fluctuations (Fig. 10.3)



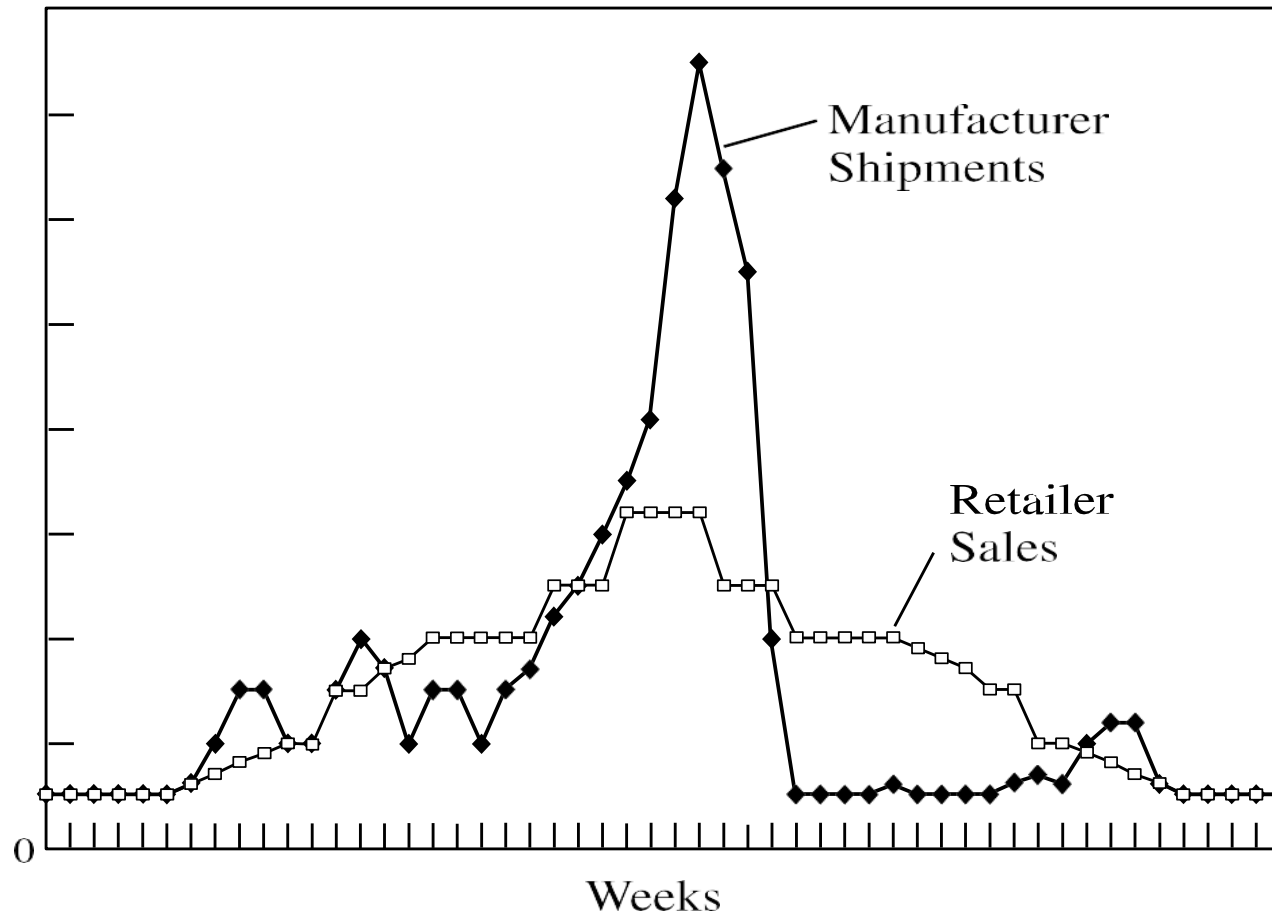
## 10.2. Obstacles to Coordination in a Supply Chain

- Demand and order stream with orders every five weeks (Figure 10.2)
  - Operational obstacle - ordering in large lots



## 10.2. Obstacles to Coordination in a Supply Chain

- Retailer sales and manufacturing shipments of soup (Figure 10.3)
  - Pricing obstacles - price fluctuations



## 10.3. Managerial Levers to Improve Coordination

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- Five categories of actions to overcome the obstacles and improve coordination in the supply chain
  - Aligning goals and incentives
  - Improving information visibility and accuracy
  - Improving operations to synchronize supply and demand
  - Designing pricing strategies to stabilize orders
  - Building strategic partnerships and trust



## 10.3. Managerial Levers to Improve Coordination

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- Managerial Actions
  - Aligning goals and incentives
    - Aligning goals across the supply chain
    - Aligning incentives across functions
    - Altering sales force incentives from sell-in to sell-through over a rolling horizon
  - Improving information visibility and accuracy
    - Sharing customer demand data
    - Implementing collaborative forecasting and planning
    - Designing single-stage control of replenishment (Vender managed inventory)



## 10.3. Managerial Levers to Improve Coordination

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- Managerial Actions
  - Improving operations to synchronize supply and demand
    - Reducing replenishment lead time
    - Reduce lot sizes
    - Rationing based on past sales and sharing information to limit gaming
  - Designing pricing strategies to stabilize orders or to coordinate
    - Moving from lot-size-based to volume-based quantity discounts
    - Stabilizing pricing
    - Pricing for coordination



## 10.3. Managerial Levers to Improve Coordination

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- Managerial Actions
  - Building strategic partnerships and trust
    - If strategic partnerships and trust are built:
      - Easier use of levers to achieve coordination
      - Better matching of supply and demand throughout the supply - chain by sharing of accurate information
      - Lower transaction costs between stages by eliminating duplicated effort (e.g., forecasting, counting, inspection)
      - Good conflict resolution
  - All parties must believe that the benefits of improved coordination are being shared equally



## 10.4. Some Practical Approaches to Improve Supply Chain Coordination

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- Continuous replenishment program (CRP)
  - The wholesaler or manufacturer replenishes the retailer warehouse regularly based on its inventory levels
- Vendor-managed inventories (VMI)\*
  - The manufacturer or supplier is responsible for all decisions regarding product inventories at the retailer
- Collaborative planning, forecasting, and replenishment (CPFR)
  - Retail event (e.g., promotions) collaboration
  - DC (distribution center) replenishment collaboration
- Collaborative part or product design



## 10.4. Some Practical Approaches to Improve Supply Chain Coordination

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- Vendor-managed inventory (vs. Retailer-managed inventory)
  - Vendor (manufacturer or supplier)
    - Plans the shipment sizes and delivery frequencies (or times) for buyers
    - Usually owns the inventory at the buyer location
    - (+): Improved production and transportation planning
  - Buyer (retailer)
    - Provides point of sale (POS) information to the vendor
    - Shares the inventory information with the vendor
    - Pays for the merchandise at the time of sale
    - Dictates the level of stock availability required
    - (+): Improved match of demand and supply w/o overstocking

